

Package #137

Develop a semitransparent thermal management system using perovskite-based materials with tunable optical properties that can simultaneously dissipate heat and allow visual monitoring of battery status

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented recurring problem 'onboard lithium battery thermal event' in aircraft/spacecraft installations: 10 NASA NTRS technical reports retrieved live at benchmark time on battery thermal-runaway behavior, calorimetry and fire safety. Titles: 'Enhancing Battery Safety with Fractional Thermal Runaway Calorimetry'; 'Practical Battery Thermal Modeling Techniques'; 'Comparative Analysis of Thermal Runaway Heat Output as a Function of Trigger Mechanism and Cell Format'. LIMITS STAMPED INLINE: technical-report literature establishes the problem class, NOT incident rates; single-source (NTRS) coverage disclosed (Art. XXI).

Mechanism: Semitransparent perovskite solar cells with optimized balance between efficiency and optical transparency

Contents:

- 01 - Executive Technology Brief
- 02 - Engineering Technology-Transfer Dossier
- 03 - Buyer Decision Card
- 04 - Evidence Summary
- 05 - Transfer Manifest

Status: ENGINEERING_DEFINITION. Not physically validated.